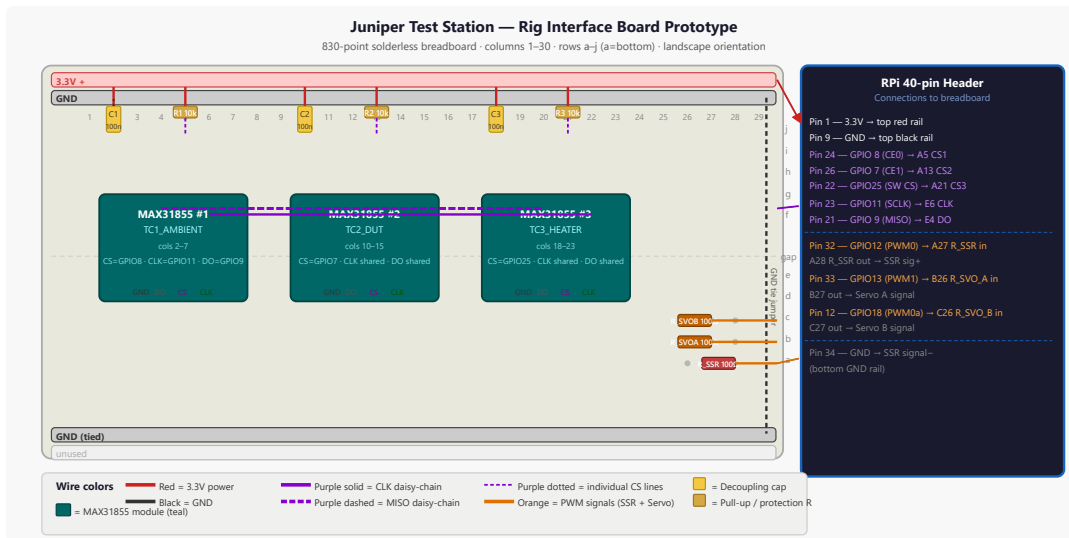


# Breadboard Prototype — Rig Interface Board

Print-ready PDF: <docs/pdf/breadboard-prototype.pdf>



The diagram above shows every component and wire at exact column/row coordinates. The text below covers placement, external connections, build order, and verification.

## Wire Color Convention

- **Red** — 3.3V power positive (MAX31855 VCC)
- **Black** — GND (all rails)
- **Orange** — PWM signal lines (heater SSR control, servo PWM)
- **Green** — digital output lines (not used on this breadboard)
- **Blue** — digital input lines (not used on this breadboard)
- **Purple** — SPI signals (MOSI, MISO, SCLK, and all CS lines)

## Coordinate System

Held in landscape orientation, long axis left-to-right:

- **Rows** are labeled `a` through `j` along the short edge — **a at the bottom**, `j` at the top. The center gap separates rows `e` (bottom half) and `f` (top half).
- **Columns** are numbered `1` through `30` along the long edge — **column 1 on the left**, column 30 on the right.
- A tie point is identified by `<row letter><column number>` — e.g., `E5` is row `e`, column 5 (lower area, left-center).
- Power rails: **top red rail** = 3.3V (outermost above row `j`); **top black rail** = GND (just below top red); **bottom red rail** = unused; **bottom black rail** = GND (tied to top GND rail at column 30).

This breadboard uses a standard 830-point full-size layout (63 column × a–j rows plus two split power rail pairs at top and bottom).

---

## Component Placement

| Component                    | Breadboard location                                                                                                        | Notes                                                                         |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| MAX31855 #1<br>(TC1_AMBIENT) | Module straddles gap, pins in rows <b>e / f</b> , columns 2–7                                                              | 8-pin SIP: VCC=E2, GND=E3, DO=E4, CS=E5, CLK=E6. Top row (f) mirrors          |
| MAX31855 #2<br>(TC2_DUT)     | Module straddles gap, pins in rows <b>e / f</b> , columns 10–15                                                            | VCC=E10, GND=E11, DO=E12, CS=E13, CLK=E14                                     |
| MAX31855 #3<br>(TC3_HEATER)  | Module straddles gap, pins in rows <b>e / f</b> , columns 18–23                                                            | VCC=E18, GND=E19, DO=E20, CS=E21, CLK=E22                                     |
| C1 (100nF ceramic)           | Vertical: top leg <b>E2</b> , bottom leg drops to top GND rail                                                             | Decoupling on TC1 VCC — non-polarised, either orientation                     |
| C2 (100nF ceramic)           | Vertical: top leg <b>E10</b> , bottom leg drops to top GND rail                                                            | Decoupling on TC2 VCC                                                         |
| C3 (100nF ceramic)           | Vertical: top leg <b>E18</b> , bottom leg drops to top GND rail                                                            | Decoupling on TC3 VCC                                                         |
| R1 (10kΩ CS pull-up, TC1)    | Horizontal: <b>A5</b> ↔ top red (3.3V) rail at column 5                                                                    | Keeps CS1 high at startup; one leg in A5, other to top + rail                 |
| R2 (10kΩ CS pull-up, TC2)    | Horizontal: <b>A13</b> ↔ top red rail at column 13                                                                         | CS2 pull-up                                                                   |
| R3 (10kΩ CS pull-up, TC3)    | Horizontal: <b>A21</b> ↔ top red rail at column 21                                                                         | CS3 pull-up                                                                   |
| R_SSR (100Ω, ¼W)             | Horizontal: <b>A27</b> ↔ <b>A28</b>                                                                                        | GPIO12 → SSR signal+ series resistor; left leg is input, right leg exits east |
| R_SVO_A (100Ω, ¼W)           | Horizontal: <b>A26</b> ↔ <b>A27</b> (shares A27 column bus with R_SSR at row <b>b</b> ) — place at <b>B26</b> ↔ <b>B27</b> | GPIO13 → Servo A PWM series resistor                                          |
| R_SVO_B (100Ω, ¼W)           | Horizontal: <b>C26</b> ↔ <b>C27</b>                                                                                        | GPIO18 → Servo B PWM series resistor                                          |

**Module pin-out note:** Standard MAX31855 breakout boards (Adafruit or clones) expose 5 pins: VCC, GND, DO (MISO), CS, CLK — in that order from left to right. Confirm your specific breakout's silkscreen before inserting. The module body straddles the center gap; pins on the bottom-half rows ( **e** ) are the ones that matter for wiring since they share column buses with rows **a – d**.

## External Connections

All wires run from the breadboard to the RPi 40-pin header (or to the off-board SSR/servo terminals) via color-coded Dupont jumpers.

### Power and Ground

| Breadboard tap       | Wire color | Destination                    | Notes                                            |
|----------------------|------------|--------------------------------|--------------------------------------------------|
| Top red rail         | Red        | RPi pin 1 (3.3V)               | Powers all three MAX31855 VCC pins + CS pull-ups |
| Top black rail       | Black      | RPi pin 9 (GND)                | Common GND for all modules                       |
| Top black rail (tie) | Black      | Bottom black rail at column 30 | Single jumper to extend GND across board         |

### SPI Bus (shared by all three MAX31855 modules)

| Breadboard tap                         | Wire color | RPi pin | BCM GPIO      | Notes                                                                               |
|----------------------------------------|------------|---------|---------------|-------------------------------------------------------------------------------------|
| E6 (CLK, TC1) + jumpers to E14 and E22 | Purple     | Pin 23  | GPIO11 (SCLK) | One wire from RPi to E6; short purple jumpers E6→E14 and E14→E22 to daisy-chain CLK |
| E4 (DO, TC1) + jumpers to E12 and E20  | Purple     | Pin 21  | GPIO9 (MISO)  | Same daisy-chain pattern for MISO                                                   |

### Chip-Select Lines (individual per module)

| Breadboard tap       | Wire color | RPi pin | BCM GPIO    | Notes                                             |
|----------------------|------------|---------|-------------|---------------------------------------------------|
| A5 (CS, TC1_AMBIENT) | Purple     | Pin 24  | GPIO8 (CE0) | Hardware SPI CS0                                  |
| A13 (CS, TC2_DUT)    | Purple     | Pin 26  | GPIO7 (CE1) | Hardware SPI CS1                                  |
| A21 (CS, TC3_HEATER) | Purple     | Pin 22  | GPIO25      | Software GPIO CS — configure as output, pull high |

## Heater SSR and Servo Outputs

| Breadboard tap              | Wire color    | RPi pin                  | BCM GPIO          | Notes                                    |
|-----------------------------|---------------|--------------------------|-------------------|------------------------------------------|
| <b>A27</b> (R_SSR input)    | <b>Orange</b> | Pin 32                   | GPIO12 (PWM0)     | GPIO12 → R_SSR (100Ω) → SSR signal+      |
| <b>A28</b> (R_SSR output)   | <b>Orange</b> | SSR terminal 3 (signal+) | —                 | Exits east from breadboard to SSR        |
| <b>B26</b> (R_SVO_A input)  | <b>Orange</b> | Pin 33                   | GPIO13 (PWM1)     | GPIO13 → R_SVO_A (100Ω) → Servo A signal |
| <b>B27</b> (R_SVO_A output) | <b>Orange</b> | Servo A signal wire      | —                 | Exits east                               |
| <b>C26</b> (R_SVO_B input)  | <b>Orange</b> | Pin 12                   | GPIO18 (PWM0 alt) | GPIO18 → R_SVO_B (100Ω) → Servo B signal |
| <b>C27</b> (R_SVO_B output) | <b>Orange</b> | Servo B signal wire      | —                 | Exits east                               |
| Bottom black rail (col 28)  | <b>Black</b>  | SSR terminal 4 (signal-) | —                 | SSR control GND                          |

## Build Order

Wire colors in **bold** below. Work left to right, power last-connect, power first-disconnect.

1. **Install C1, C2, C3 first** (anchor the decoupling before the ICs go in). Push each 100nF ceramic cap vertically: top leg into **E2, E10**, and **E18** respectively; bottom leg drops straight down to the top black (GND) rail. Non-polarised — either orientation. Verify no bridges with adjacent columns.
2. **Insert MAX31855 #1** so VCC is at **E2**, GND at **E3**, DO at **E4**, CS at **E5**, CLK at **E6**. Module body straddles the center gap. Double-check the silkscreen — DO NOT insert backwards. Confirm C1's VCC leg (E2) is on the same column bus as the module's VCC pin.
3. **Insert MAX31855 #2** at columns 10–14 (VCC=E10, GND=E11, DO=E12, CS=E13, CLK=E14). Confirm C2 at E10.
4. **Insert MAX31855 #3** at columns 18–22 (VCC=E18, GND=E19, DO=E20, CS=E21, CLK=E22). Confirm C3 at E18.
5. **Install CS pull-up resistors R1, R2, R3**. Each is 10kΩ. R1: one leg into **A5**, the other leg into the top red rail at column 5. R2: **A13** to top red rail at column 13. R3: **A21** to top red rail at column 21. These hold CS lines HIGH when the RPi is booting or the pin is not yet configured — prevents false SPI transactions.
6. **Wire the SPI bus daisy-chain**.

7. **Purple** jumper: CLK from **E6** (TC1) → **E14** (TC2). Another **purple** jumper: E14 → **E22** (TC3). CLK is now shared.
  8. **Purple** jumper: DO (MISO) from **E4** → **E12**. Another **purple** jumper: E12 → **E20**. MISO is now shared.
  9. **Install protection resistors R\_SSR, R\_SVO\_A, R\_SVO\_B** in the columns 26–28 area.
  10. R\_SSR (100Ω) horizontal at **A27** ↔ **A28**.
  11. R\_SVO\_A (100Ω) horizontal at **B26** ↔ **B27**.
  12. R\_SVO\_B (100Ω) horizontal at **C26** ↔ **C27**.
  13. **Connect power rails**. Do NOT connect RPi yet.
  14. **Black** jumper: top black rail → bottom black rail at column 30 (GND tie across full board).
  15. Verify no shorts across + and - rails with a DMM (continuity mode). Should read open.
  16. **Connect RPi power and GND**. With the RPi powered OFF:
  17. **Red** wire: RPi pin 1 (3.3V) → top red rail at column 1.
  18. **Black** wire: RPi pin 9 (GND) → top black rail at column 1. Power on RPi. Measure top red rail = 3.3V, top black rail = 0V. Check E2/E10/E18 all read 3.3V.
  19. **Connect all signal wires** per the External Connections tables above. RPi SPI pins last. Clip each Dupont female onto the corresponding RPi pin one at a time, confirming pin number against the 40-pin header diagram before pushing:
    - **Purple** × 3: CS1 (pin 24), CS2 (pin 26), CS3 (pin 22) → A5, A13, A21
    - **Purple** × 2: SCLK (pin 23) → E6; MISO (pin 21) → E4
    - **Orange** × 3: GPIO12 (pin 32) → A27; GPIO13 (pin 33) → B26; GPIO18 (pin 12) → C26
    - **Orange** × 2: A28 → SSR terminal 3; B27 → Servo A signal; C27 → Servo B signal
    - **Black**: bottom black rail → SSR terminal 4 (signal GND)
-

# Verification Checklist

| Stage                     | What to check                                                                                                  | How to check                                                                           |
|---------------------------|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| 1 — Power rail            | Top red rail = 3.3V, GND rail = 0V                                                                             | DMM, probes on rail strips                                                             |
| 2 — VCC at modules        | E2, E10, E18 all = 3.3V                                                                                        | DMM tip on each tie point                                                              |
| 3 — GND at modules        | E3, E11, E19 all = 0V                                                                                          | DMM                                                                                    |
| 4 — CS pull-ups           | A5, A13, A21 all = 3.3V with RPi booted (before Python runs)                                                   | DMM — confirms R1/R2/R3 wired to + rail                                                |
| 5 — SPI CLK daisy         | Probe E6, E14, E22 simultaneously while running <code>spidev</code> test — all should show same clock waveform | Oscilloscope or logic analyser; or verify each column with DMM in AC mode              |
| 6 — MISO continuity       | Measure resistance E4→E12→E20 with RPi off — should be <5Ω                                                     | DMM continuity across the daisy-chain                                                  |
| 7 — TC1 reads             | <code>python3 -c "import board, busio, adafruit_max31855; ..."</code> returns ambient temp ±5°C                | Software test — see <code>software/test_tc.py</code>                                   |
| 8 — TC2, TC3 read         | Same test for GPIO7 (CS2) and GPIO25 (CS3)                                                                     | Software                                                                               |
| 9 — SSR signal            | RPi GPIO12 = 3.3V with <code>gpio write 12 1</code> (WiringPi) or <code>raspi-gpio set 12 op dh</code>         | DMM at A28 (R_SSR output) reads 3.3V                                                   |
| 10 — Servo signal         | GPIO13 and GPIO18 toggle visible on DMM/scope at A28 column area                                               | DMM in frequency mode at B27 / C27 output                                              |
| 11 — No cross-rail shorts | Power off, DMM continuity: top red rail ↔ top black rail → should be OPEN                                      | Any continuity beep = module or component bridge — locate and fix before next power-on |